WORKED SOLUTION

STEP I 1988, Question 11

Mechanics • Centre of mass

STEP I 1988, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q11

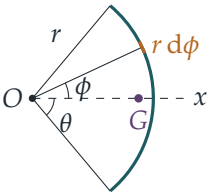
The key idea

For the arc, symmetry puts the centre of mass on the bisecting radius, and one integral over the angle gives its distance from the centre. For the framework, the striking fact is that it balances on *every* point of the semicircle. The floor's reaction always acts along a radius of the semicircle, and the only point lying on all those radii is the centre. So the centre of mass of the whole framework must be the centre of the semicircle, and taking moments about that point gives an equation for h.

Solution

The uniform arc

Let the arc have centre O and density ρ per unit length. Measure the angle ϕ from the bisecting radius, so the arc is $-\theta \le \phi \le \theta$. Put the x-axis along the bisector. By symmetry in that axis the centre of mass lies on it, so we only need $\bar{x}$.



The element between ϕ and $\phi + \mathrm{d}\phi$ has length $r\,\mathrm{d}\phi$, so mass $\rho r\,\mathrm{d}\phi$, and its x-coordinate is $r\cos\phi$. The total mass is $\rho r \cdot 2\theta$. Hence

$$\bar{x} = \frac{\displaystyle\int_{-\theta}^{\theta} r\cos\phi\ \rho r\,\mathrm{d}\phi}{2\rho r\theta} = \frac{r[\sin\phi]_{-\theta}^{\theta}}{2\theta} = \frac{r\sin\theta}{\theta}.$$

So the centre of mass lies on the bisecting radius, at distance

$$\boxed{\bar{x} = \frac{r\sin\theta}{\theta}}$$

from the centre. For a semicircular arc, $\theta = \frac{\pi}{2}$, and this is $\dfrac{2r}{\pi}$, which is the case we need next.

The framework: where must the centre of mass be?

We model the framework as rigid and made of uniform thin rods of the same density ρ per unit length, and the floor as horizontal. Let O be the centre of the semicircle, which is the midpoint of the bottom side of the rectangle.

Ask yourself Why does "equilibrium at any point of the semicircle" pin down the centre of mass completely?

Suppose the framework rests with the point P of the semicircle on the floor. The floor is tangent to the semicircle at P, so the vertical through P is perpendicular to that tangent: it is the radius line OP. Only two forces act, the weight at the centre of mass G and the reaction at P. For equilibrium they must act along the same line, so G lies on the vertical through P, which is the line OP. This holds for every P on the semicircle. Two different points P give two different lines through O, and these meet only at O. Hence $G = O$.

Conversely, if $G = O$ then for every P the weight acts along OP, straight through the point of contact, so the framework is in equilibrium wherever it touches the floor. The condition is therefore exactly "G is at O".

Moments about the bottom side

Take moments about the line of the bottom side, measuring heights upwards from O (so the semicircle is below the line). The pieces are:

piece	mass	height of its centre	moment
top rod	$2r\rho$	$2h$	$4rh\rho$
two side rods	$2 \times 2h\rho$	h	$4h^2\rho$
bottom rod	$2r\rho$	0	0
semicircle	$\pi r\rho$	$-\dfrac{2r}{\pi}$	$-2r^2\rho$

The semicircle's centre of mass comes from the arc formula with $\theta = \frac{\pi}{2}$, and it is *below* O. Setting the total moment to zero, because G is at height 0, and dividing by 2ρ:

$$2h^2 + 2rh - r^2 = 0.$$

This is a quadratic in h, with roots

$$h = \frac{-2r \pm \sqrt{4r^2 + 8r^2}}{4} = \frac{(-1 \pm \sqrt{3})\, r}{2}.$$

Since $h > 0$ we take the positive root:

$$\boxed{h = \frac{(\sqrt{3}-1)\,r}{2} \approx 0.366\,r}$$

Notice that the bottom rod has zero moment about its own line. The answer is the same whether or not the framework includes it, so a slightly different reading of the diagram doesn't matter.

Checks

- As $\theta \to 0$, $\frac{r\sin\theta}{\theta} \to r$: a very short arc has its centre of mass on the arc itself. For a full circle, $\theta = \pi$, we get 0, the centre, as symmetry demands.
- With $h = \frac{(\sqrt{3}-1)r}{2}$: $2h^2 = (2-\sqrt{3})r^2$ and $2rh = (\sqrt{3}-1)r^2$, which add to r^2 as required.
- We computed the centre of mass numerically by splitting every rod into 20 000 pieces. With $r = 1$ and $h \approx 0.36603$, it lies within 10^{-9} of O. Increasing h by 10% moves it up by about 0.03, so the framework would no longer balance.

Common slips

- Using $\frac{4r}{3\pi}$, which is for a semicircular *lamina*, instead of $\frac{2r}{\pi}$ for a semicircular *wire*.
- Weighting by the angle rather than by length: the mass of the element is $\rho r\,\mathrm{d}\phi$, and the factor r cancels only because it also appears in the total mass.
- Asserting that G is at O without a reason. The argument is that the reaction always acts along a radius.

Beyond STEP

The arc integral is the simplest example of a centre of mass as a *line integral* $\int \mathbf{r}\,\rho\,\mathrm{d}s$ along a curve. First-year courses go on to surfaces and solids using double and triple integrals, and Pappus's theorems link these centres of mass to areas and volumes of revolution. The balancing argument is also the idea behind rocking stability, where a body resting on a curved surface is stable if its centre of mass lies below the centre of curvature. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats centres of mass and rigid-body equilibrium in this spirit.

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